Automated Irregular Shaped Stone Stacking via Pose Optimization and Monte Carlo Tree Search

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***Abstract*—We study automated stacking of irregular objects to approximate prescribed target shapes without support from container walls. Our framework combines bilevel pose optimization with batched, receding-horizon Monte Carlo Tree Search (MCTS) over object choices and placement poses. The optimizer couples geometric objectives with contact-force optimization to evaluate force and torque equilibrium across the structure under friction constraints, using a bodywise decomposition for GPU parallelization. Physical simulation validates placements and supplies settled successor states for multi-step search. We hypothesize that combining structural stability evaluation with lookahead improves target coverage while maintaining stability under a fixed planning budget.**

I. Introduction

Automated packing and palletizing have long been studied to improve space utilization and material handling in logistics and manufacturing [6]. Recent robotic packing methods accommodate irregular objects while considering packing density, stability, and placement feasibility [22, 9]. Constructing freestanding structures from natural stones requires finding mutually supporting arrangements among uneven surfaces while achieving a prescribed target shape without support from container walls.

Several stone-stacking methods use placement criteria developed for particular construction settings, including tower-building objectives [7] and masonry-inspired geometric rules [12, 19]. For example, Johns et al.'s double-faced wall planner generates candidates along the intersection of the target wall surfaces and the existing structure, and applies inward-slope heuristics tailored to wall construction [12]. The ability to construct freeform walls [13] demonstrates flexibility in wall geometry, while extending beyond such construction settings motivates planning driven by target geometry without wall-specific placement rules. Structural stability also requires reasoning beyond the support of the newly placed object: its load can affect the equilibrium of objects beneath it. Liu and Napp [17] explicitly address structural strength during planning, demonstrating the value of evaluating the structure as a whole. However, assessing the current structure alone does not determine how a placement affects future support and space utilization. Menezes et al. [20] address sequential placement through learned action values that account for future rewards, with a prescribed object sequence and fixed orientations. We investigate explicit lookahead over object choices and placement poses, coupled with structural stability evaluation and target-shape objectives.

We consider stacking a finite inventory of n irregular objects to approximate a target shape $T \subset \mathbb{R}^3$. Let $O_i(q_i) \subset \mathbb{R}^3$ denote the volume occupied by object i at pose $q_i \in \mathrm{SE}(3)$. The final configuration objective is

$$\min_{q_1,\cdots,q_n \in \mathrm{SE}(3)} \mathbb{V}\left(T \setminus \bigcup_{i=1}^{n} O_i(q_i)\right) \quad \text{s.t.} \quad c_{\text{phys}}(O_1(q_1),\cdots,O_n(q_n)) = 0, \tag{1}$$

where $\mathbb{V}(\cdot)$ denotes volume and $c_{\text{phys}} = 0$ represents stability and non-interpenetration constraints. Jointly optimizing all object poses under coupled physical constraints is computationally demanding. To obtain a practical planning procedure, we decompose construction into sequential decisions, selecting one object and optimizing its placement at each step while checking the resulting structure's physical feasibility. This decomposition reduces the dimension of each pose optimization, but each placement changes the support geometry, free space, and remaining inventory. Multi-step lookahead allows the planner to evaluate these future consequences beyond immediate placement quality. Our central question is how to combine this lookahead with structural stability evaluation in a common framework for different target shapes.

We address this question with a framework coupling bilevel pose optimization and Monte Carlo Tree Search (MCTS). The pose optimizer refines geometric proposals using target-shape objectives and a physical objective that evaluates force and torque equilibrium across the structure under friction constraints. The target enters through its geometric representation, allowing the same optimization formulation to be applied to different target shapes. A bodywise ADMM decomposition enables parallel solution of the contact-force subproblem.



Fig. 1: Preliminary stacking results. Left: simulated structures for different target shapes. Right: a physical stone structure from an outdoor experiment.

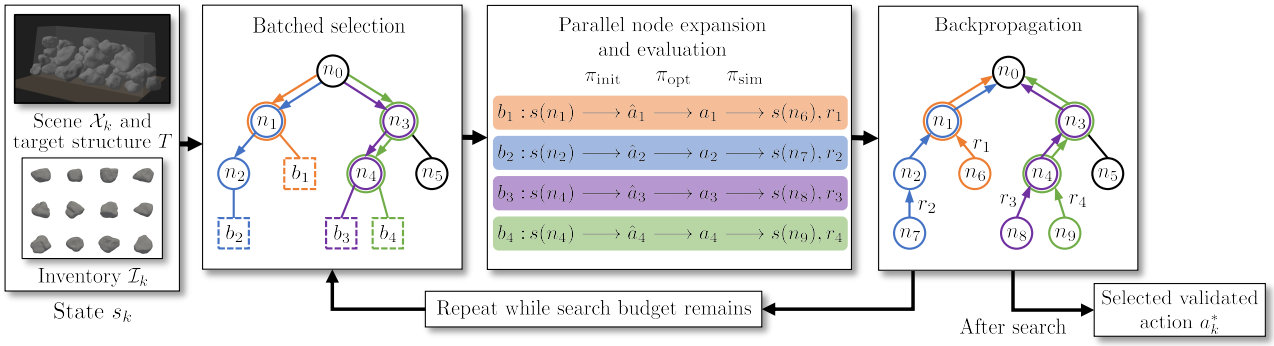


Fig. 2: Batched MCTS sequence planning. Colors identify expansion requests b_j and their tree paths. Requests are evaluated in parallel through π_{init}, π_{opt}, and π_{sim}, and returns are backpropagated along the selected paths. Search repeats within the budget before selecting a physically validated root action a_k^*.

The sequence planner embeds pose refinement and physical simulation in batched, receding-horizon MCTS, following prior work on batched simulation-based search [8]. Simulation supplies settled successor states and rejects unstable placements, while multi-step lookahead compares their effects on subsequent construction.

We hypothesize that combining structural stability evaluation with multi-step lookahead improves target coverage while maintaining stability across different target geometries under a fixed planning budget. Comparisons with greedy placement and candidate-only stability evaluation will test the effects of lookahead and structural coupling, respectively. Evaluations across target shapes and search horizons will examine the applicability of the common formulation and its computational tradeoffs. The proposed contributions are:

- A bilevel placement optimizer that combines explicit target geometry with contact-force optimization across the structure, using bodywise ADMM updates for GPU computation.
- A batched, receding-horizon stacking planner that integrates pose refinement and physical simulation with multi-step lookahead over object choices and placement poses.

II. Related Works

A. Automated Stone Stacking

Furrer et al. [7] combined gradient-based pose search with physics simulation for stone stacking. Johns et al. [12] integrated perception, planning, and excavator manipulation for wall construction, later using constrained registration and signed-distance-field classification for prescribed wall geometries [13]. Liu et al. [19] selected simulation-generated poses using hierarchical filters informed by masonry heuristics. Liu and Napp [17] incorporated structural strength evaluation into sequence planning with physical validation, later introducing stackability measures for corbelled shelter construction [18]. Menezes et al. [20] used deep Q-learning on elevation maps to select planar placements, rewarding target occupancy and penalizing object displacement. Their fixed-sequence, fixed-orientation construction strategies inform our initial pose sampler.

B. Irregular Object Packing

Wang and Hauser [22] packed nonconvex objects through heightmap minimization, considering stability, robot motion feasibility, and replanning for execution errors. Huang et al. [9] jointly planned object order and placement through hierarchical reinforcement learning with multi-view heightmaps and compactness, pyramidality, and simulated-stability rewards.

C. Contact-Aware Pose Optimization

Lee et al. [16] use differentiable Contact Factor Graphs for gradient-based manipulation planning. BODex [5] couples lower-level contact-force optimization with upper-level grasp synthesis, while GraspQP [23] defines a differentiable force-closure energy through a quadratic program. Our optimizer couples candidate placement with structure-wide equilibrium, using bodywise force decomposition for parallel evaluation.

D. Simulation-Based Sequential Manipulation Planning

King et al. [14] extended MCTS to nonprehensile rearrangement under uncertainty using learned and heuristic rollout policies. Huang et al. [8] introduced Parallel Monte Carlo Tree Search with Batched Simulations (PMBS), coupling parallel search with GPU rigid-body simulation for object retrieval from clutter. Our stacking planner adopts batched evaluation, combining geometric sampling, stability-aware pose refinement, and structural simulation at each expansion.

III. Overview and Preliminaries

A. Shape Representation

We represent the target T and each object O_i as unions of smooth convex bodies modeled by differentiable support functions (DSFs) [15]. For a compact convex body $C \subset \mathbb{R}^3$, its support function is $h_C(x) = \max_{p \in C} x^\top p$ for $x \neq 0$. At pose $q = (R, t) \in \mathrm{SE}(3)$, we write

$$C(q) = \{Rp + t \mid p \in C\}, \qquad O_i(q_i) = \bigcup_j C_{ij}(q_i),$$

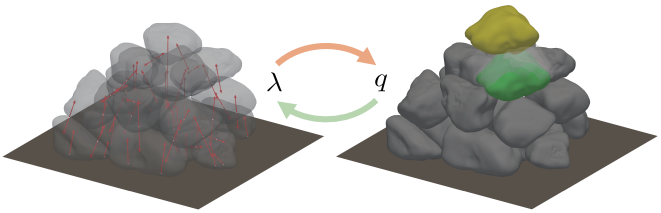


Fig. 3: Bilevel pose optimization. Left: optimizing contact forces λ under friction constraints minimizes structural equilibrium residuals at fixed q. Right: refining $q \in \mathrm{SE}(3)$ minimizes the geometric and optimized physical objectives.

where C_{ij} are the convex components in object i's local frame. Similarly, $T = \bigcup_j T_j$, where T_j are fixed convex components in the world frame. DSFs support differentiable signed-distance and contact-geometry computations between posed convex bodies [2], enabling gradient-based pose optimization.

B. Markov Decision Process Formulation

To address the construction objective in Eq. (1), we model stacking as a finite-horizon sequential decision process in which one object is placed at each step. Let $\mathcal{X}_k = \{(i, q_i) \mid i \in \mathcal{P}_k\}$ be the current scene and $\mathcal{I}_k = \{1, \ldots, n\} \setminus \mathcal{P}_k$ the remaining inventory. The state, action, and deterministic transition function are

$$s_k = (\mathcal{X}_k, \mathcal{I}_k, T), \quad a_k = (i_{k+1}, q_{k+1}), \quad s_{k+1} = F(s_k, a_k),$$

where $i_{k+1} \in \mathcal{I}_k$. The transition inserts the proposed pose into the scene, simulates the resulting structure, and returns the settled scene with $\mathcal{I}_{k+1} = \mathcal{I}_k \setminus \{i_{k+1}\}$. The reward $r(s_k, a_k)$ scores target-contained volume and physical quality; feasibility checks separately reject invalid transitions (Sec. V). The sequence planner therefore operates on the MDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, F, r, \gamma)$.

C. Monte Carlo Tree Search

Monte Carlo Tree Search (MCTS) incrementally constructs a search tree $\mathcal{T} = (\mathcal{N}, \mathcal{E})$ over the MDP. Each node $n \in \mathcal{N}$ represents a state $s(n)$, and each edge $(n, a) \in \mathcal{E}$ represents an action and its transition to a child node. Let $N(n)$ denote the number of completed evaluations and $Q(n)$ the empirical mean of discounted returns backpropagated to that node.

Each iteration comprises four operations. During *selection*, the tree policy follows promising children by balancing empirical value and exploration. During *expansion*, a new action is generated at the selected node; *evaluation* obtains its reward and successor state. A nonterminal leaf may also receive a value estimate $\hat{V}(s)$ from a rollout or problem-specific heuristic. Finally, *backpropagation* propagates the discounted return along the selected path to update $Q(n)$ and $N(n)$.

Here, actions place remaining objects, with evaluation involving pose refinement and physical simulation. Our geometry-informed proposals, batched selection, and finite-horizon leaf evaluation are described in Sec. V.

D. Overall Structure of the Sequence Planner

Figure 2 illustrates the MCTS sequence planner. For each expansion, the initial pose sampler π_{init} proposes a candidate action $\hat{a}_k = (i_{k+1}, \hat{q}_{k+1})$, specifying both the object index and its initial pose. The pose optimization module π_{opt} refines the pose to obtain $a_k = (i_{k+1}, q_{k+1})$. The simulation module π_{sim} evaluates the placement and its effect on the structure, producing the settled successor state before disturbance trials. Validity and simulation statistics determine tree admission and reward, giving the execution flow

$$s_k \xrightarrow{\pi_{\text{init}}} \hat{a}_k \xrightarrow{\pi_{\text{opt}}} a_k \xrightarrow{\pi_{\text{sim}}} s_{k+1}, \; r(s_k, a_k).$$

IV. Pose Optimization

The module π_{opt} refines $\hat{q}_{k+1}$ from π_{init} toward a stable, physically feasible placement q_{k+1}.

A. Single Object Pose Optimization Problem

Given s_k, we initialize q at $\hat{q}_{k+1}$ and solve the bilevel problem

$$\begin{aligned} \min_{q} \quad & \left[f_{\text{geo}}(q) + \min_{\lambda} f_{\text{phys}}(q, \lambda) \right] \\ \text{s.t.} \quad & g_{\text{cone},j}(q, \lambda_j) \le 0, \quad j = 1, \cdots, n_c \end{aligned} \tag{2}$$

where $q \in \mathrm{SE}(3)$ is the candidate object's pose and $\lambda = (\lambda_1, \cdots, \lambda_{n_c}) \in \mathbb{R}^{3n_c}$ collects all contact forces in the resulting structure, including inter-object and ground contacts. This formulation follows prior contact-aware optimization methods [16, 5].

The geometric objective f_{geo} is designed as follows:

$$\begin{aligned} f_{\text{geo}}(q) = w_{\text{h}} f_{\text{h}}(q) + w_{\text{in}} f_{\text{in}}(q \mid T, O_{i_{k+1}}) \\ + w_{\text{pen}} f_{\text{pen}}(q \mid \mathcal{X}_k, O_{i_{k+1}}) \end{aligned}$$

with positive weights. For $q = (R, t)$, $f_{\text{h}}(q) = t_z$ is the world-frame height of the candidate's pose origin. Writing $O = O_{i_{k+1}}$, we penalize proximity to existing objects and the fixed ground, indexed by i_0, as

$$f_{\text{pen}}(q) = \sum_{j \in \mathcal{P}_k \cup \{i_0\}} \left[\epsilon_{\text{margin}} - \mathrm{d}(O(q), O_j(q_j)) \right]_+^2,$$

where d is signed distance (positive for separation), $[x]_+ = \max(x, 0)$, and $\epsilon_{\text{margin}} \ge 0$ is the clearance margin. With $h_{O(q)}(u) = \max_j h_{C_{i_{k+1},j}(q)}(u)$, we define

$$\begin{aligned} m_{\text{in}}(q) &= \max_j \min_{u \in \mathbb{S}^2} \left[h_{T_j}(u) - h_{O(q)}(u) \right], \\ f_{\text{in}}(q) &= \left[-m_{\text{in}}(q) \right]_+^2. \end{aligned}$$

This term penalizes violations of containment of the entire object within a single convex target component, a conservative condition for containment in T. It is differentiable almost everywhere in q; we compute pose derivatives at regular inner solutions by implicit differentiation. Nonsmooth transitions have not caused practical difficulties in our implementation.

Let $\mathcal{P}_{k+1} := \mathcal{P}_k \cup \{i_{k+1}\}$, excluding the ground index i_0. For distinct $l, m \in \mathcal{P}_{k+1} \cup \{i_0\}$, let $I_{l,m} \subset \{1, \ldots, n_c\}$ index contacts between bodies l and m. Each object's contact

set $I_l := \bigcup_{m\in(\mathcal{P}_{k+1}\cup\{i_0\})\setminus\{l\}} I_{l,m}$ includes ground contacts, but the physical objective excludes the ground's equilibrium residual:

$$f_{\text{phys}}(q,\lambda) = w_{\text{eq}} \sum_{l\in\mathcal{P}_{k+1}} \|J_{c,I_l}^{\top}(q)\lambda_{I_l} + \lambda_{\text{ext},l}\|^2 + w_{\text{cp}} \sum_{j\in I_{i_{k+1}}} \mathrm{d}_j(q)^2\|\lambda_j\|^2 + \rho\sum_{j=1}^{n_c}\|\lambda_j\|^2 \tag{3}$$

The three terms respectively penalize violations of static equilibrium, contact complementarity between the candidate and the current scene, and large contact forces through regularization. Here, $\mathrm{d}_j(q)$ is the signed distance between the two bodies associated with contact candidate j. J_{c,I_l} is the contact Jacobian for object O_l, $\lambda_{\text{ext},l}$ is its external wrench, λ_{I_l} is the vector of contact forces at the contact points in I_l, and $w_{\text{eq}}, w_{\text{cp}}, \rho > 0$ are weighting and regularization parameters.

Finally, each $g_{\text{cone},j}(\cdot)$ is the smoothed friction cone constraints for the j-th contact point defined by

$$g_{\text{cone},j}(q,\lambda_j) = \sqrt{\|J_{c,j}^t\lambda_j\|^2 + \epsilon} - \mu_j J_{c,j}^n\lambda_j \tag{4}$$

where $\epsilon > 0$ is a smoothing parameter, μ_j is the friction coefficient, and $J_{c,j}^t$ and $J_{c,j}^n$ are the tangential and normal components of the Jacobian of the contact constraint.

B. Upper-Level Problem Optimization on SE(3) Manifold

Denoting $\lambda^*(q)$ as the optimal λ of the lower-level problem for a given pose q, we define the force dependent objective function as $f_{\text{phys}}(q) := f_{\text{phys}}(q,\lambda^*(q))$. Then, using the *Envelope theorem* [4], we can compute the gradient of $f_{\text{phys}}(q)$ with respect to q as follows:

$$\frac{df_{\text{phys}}(q)}{dq} = \frac{\partial f_{\text{phys}}(q,\lambda^*)}{\partial q} + \sum_{j=1}^{n_c}\nu_j^*\frac{\partial g_{\text{cone},j}(q,\lambda_j^*)}{\partial q}$$

where ν_j^* is the optimal dual variable associated with the j-th friction cone constraint $g_{\text{cone},j}(q,\lambda_j^*) \le 0$. Using the above gradient, we optimize the placement pose q on SE(3) manifold based on the Riemannian trust-region method [1] combined with BFGS Hessian approximation [10]. We progressively adjust the geometric and physical objective weights and resolve Eq. (2) to guide the local search.

C. Lower-Level Problem Optimization with ADMM

For a fixed q, the lower-level problem is a convex optimization problem with respect to λ which can be efficiently solved. Moreover, exploiting the inherent structure of the lower-level problem, we can reformulate it to be separable with respect to the objects $\mathcal{P}_{k+1}$ in the scene and apply the alternating direction method of multipliers (ADMM) [3] to solve it in a scalable manner. For each contact force λ_j, we introduce two local copies $\lambda_{j,1}$ and $\lambda_{j,2}$ associated with the two contacting bodies, including the fixed ground. Specifically, for $j \in I_{l,m}$ with $l < m$, the copies are associated with bodies l and m, respectively. We then impose the consensus constraints $\lambda_{j,1} = \lambda_{j,2}$. Collecting all local copies into $\tilde{\lambda}$ makes the force and torque equilibrium term separable by body: each body's residual depends only on its own local contact-force variables. The consensus constraints maintain agreement between copies of each shared contact force. The lower-level problem can therefore be reformulated as

$$\min_{\tilde{\lambda},z} \quad \tilde{f}_{\text{phys}}(q,\tilde{\lambda}) + \sum_{\substack{j=1,\dots,n_c\\ s=1,2}} \delta_{K_j}(z_{j,s})$$
$$\text{s.t.} \quad \lambda_{j,1} = \lambda_{j,2} = z_{j,1} = z_{j,2}, \qquad j = 1,\dots,n_c.$$

where $\tilde{f}_{\text{phys}}(q,\tilde{\lambda})$ is the extended physical force function and δ_{K_j} is the indicator function of the convex set of feasible contact forces $K_j := \{\lambda \in \mathbb{R}^3 \mid g_{\text{cone},j}(q,\lambda) \le 0\}$. Here, $z_{j,s}$ is an auxiliary variable associated with the j-th contact point and $s \in \{1,2\}$. At each ADMM iteration, the objectwise-separable subproblem decomposes into $|\mathcal{P}_{k+1}|$ independent subproblems, one for each object O_l. An equivalent four-dimensional second-order cone (SOC) lifting enables closed-form cone projections for the friction updates, while consensus updates reduce to averaging.

The pose optimization module is implemented using CUDA to leverage GPU parallelization for efficient computation of the gradients and the ADMM updates.

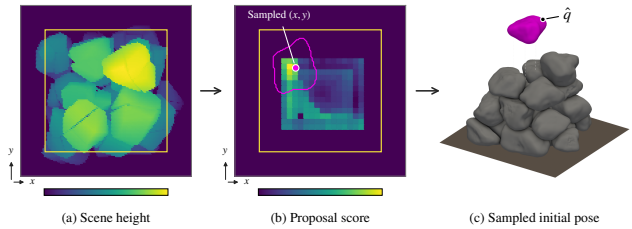


Fig. 4: Initial pose sampling for a fixed object and orientation: (a) scene height, (b) proposal score and sampled planar location, and (c) initial pose $\hat{q}$ above the scene. Magenta links the sampled location and candidate object.

V. Sequence Planning

A. Initial Pose Sampling

The sampler π_{init} allocates fresh proposals to remaining objects in proportion to their volumes and considers a finite set of orientation hypotheses for each object. It prioritizes connected growth on the lowest incomplete layer, seeding an empty scene near feasible target corners. For an object i, orientation R, and feasible planar location $\mathbf{x}$, the geometric score is

$$\sigma(\mathbf{x} \mid i, R) = w_{\text{s}}\sigma_{\text{s}} + w_{\text{f}}\sigma_{\text{f}} + w_{\text{b}}\sigma_{\text{b}} - w_{\text{h}}\sigma_{\text{h}},$$

with positive weights. The terms measure the fraction of bottom samples near predicted contact, unfilled layer area beneath the footprint, proximity to the feasible placement boundary, and normalized mean contact height, respectively. The floor and boundary preferences are inspired by Menezes et al. [20]. Figure 4 illustrates the resulting proposal score and sampled initial pose.

Within eligible pools, the standard sampler combines elite selection with draws without replacement weighted by $\exp(\sigma/\tau)$, $\tau > 0$. Each selected hypothesis is initialized above overlapping scene geometry with a clearance, then refined by π_{opt} and filtered for feasibility. At the search root, population refinement additionally mixes fresh proposals with planar-position and yaw mutations of promising candidates, followed by renewed pose optimization.

B. Simulation Module

The module π_{sim} first settles the new object for a fixed duration with the existing scene static, then simulates the structure. Exploration uses short evaluations that may freeze distant objects while preserving the candidate's support chain. Root placements and final continuation paths receive longer validation without distance-based freezing; explicitly fixed bodies remain static.

Structural settling requires a minimum duration and sustained low mass-normalized kinetic energy. Excessive accumulated motion, invalid contacts, or timeout without settling rejects the placement. We store the settled scene as the successor $s' = F(s, a)$. Separate trials start from this scene and apply Gaussian velocity disturbances to the new object, measuring motion of all dynamic bodies. These trials supply robustness scores but do not alter s'. Simulation uses CUDA subADMM [11].

C. Reward Design

Let q_i' be object i's pose in the settled successor s' returned by π_{sim} (Sec. V-B). We define the target-volume and object-volume ratios

$$\rho_T = \frac{\mathbb{V}\big(O_i(q_i') \cap T\big)}{\mathbb{V}(T)}, \quad \rho_O = \frac{\mathbb{V}\big(O_i(q_i') \cap T\big)}{\mathbb{V}(O_i)}.$$

The former measures the new object's contribution to filling the target; the latter measures its containment. Intersection volumes are estimated on a voxel grid. With fixed normalization scales absorbed into positive weights, the standard excavator reward is

$$\begin{aligned} r(s,a) = {} & w_T \rho_T + w_O[\rho_O - \tfrac{1}{2}]_+ - w_E e(a) \\ & - w_H\big(\max_{\Omega} H_{s'} - \langle H_{s'} \rangle_{\Omega}\big) - w_D \widetilde{D}, \end{aligned}$$

where $[x]_+ = \max(x, 0)$, so the containment bonus is active above 50%. Coverage lost through motion of earlier objects is not subtracted. $H_{s'}$ is surface height over the target footprint Ω, with zero at empty cells; $\langle\cdot\rangle_{\Omega}$ is its spatial mean.

The equilibrium score uses residual forces $\mathbf{f}_b$ and torques $\boldsymbol{\tau}_b$ returned by pose optimization:

$$e(a) = \min\left\{1, \max_b \left(\frac{\|\mathbf{f}_b\|}{m_b g} + \frac{\|\boldsymbol{\tau}_b\|}{m_b g \mathbb{V}(O_b)^{1/3}}\right)\right\}.$$

Here b indexes participating bodies and g is gravitational acceleration. For disturbance trial j, dynamic body b, and time t, define

$$D = \max_{j,b,t} \left[\|p_{j,b}(t) - p_b^0\| + \lambda_\theta d_{\text{SO(3)}}(R_{j,b}(t), R_b^0)\right],$$

where (p_b^0, R_b^0) is body b's pose in the same successor s', $d_{\text{SO(3)}}$ is angular distance, and λ_θ scales rotation. The capped excursion $\widetilde{D}$ measures the worst transient motion across trials and dynamic bodies. Physical rejection criteria apply independently of the reward.

D. Monte Carlo Tree Search

We integrate these modules in batched, receding-horizon MCTS (Fig. 2), following the parallel evaluation approach of [8].

Batched Selection. For an optimized action $a = (i, q)$, let ρ_O^{pre} denote the object-volume ratio evaluated at q, before simulation. The logit $\ell(a) = 2(\rho_O^{\text{pre}} - e(a))$ defines $P(n_p, a)$ through a softmax over selectable children; root refinement adds the normalized proposal score to ℓ for ranking. Writing $N_n = N(n)$ and similarly for Q_n and $\hat{N}_n$, an evaluated child n of n_p is selected using PUCT [21]:

$$\text{PUCT}(n) = \frac{N_n Q_n}{N_n + \hat{N}_n} + cP(n_p, a)\frac{\sqrt{\max\{N_{n_p} + \hat{N}_{n_p}, 1\}}}{1 + N_n + \hat{N}_n}, \tag{5}$$

where $c > 0$ controls exploration, $\hat{N}$ counts temporary reservations, and the first term is zero when its denominator is zero. Non-root progressive widening permits $K(n) = \min\{16, \lceil 3\max(N(n), 1)^{1/2}\rceil\}$ children based on completed visits; root capacity is set separately. Selection reserves pending actions or new proposal slots, prioritizing available pending actions before PUCT descent through evaluated children. Reservations prevent duplicate work and do not increase the widening allowance.

Parallel Expansion and Evaluation. After batch selection, fresh proposals pass through $\pi_{\text{init}} \to \pi_{\text{opt}} \to \pi_{\text{sim}}$, whereas cached optimized actions enter at simulation. Independent candidates are processed in parallel; selection never descends through an action without an evaluated successor. Failed candidates are excluded from further selection.

Leaf Value Estimation and Backpropagation. We replace additional rollouts with a discounted fixed per-step reward heuristic over a capped horizon, limited by search depth and remaining inventory. Terminal and failed leaves receive no bootstrap. Discounted returns update Q and N along selected paths, with a failure reward propagated to ancestors of rejected candidates. Temporary reservations are then cleared.

Action Selection and Tree Reuse. Among root actions with continuations validated by longer simulations, we maximize $J(a) = (1 - \beta)Q(n_a) + \beta G_{\text{val}}(a)$, where $\beta \in [0, 1]$ and G_{val} is the validated path return, including the current placement. If bounded repair finds no such path, we fall back to a physically validated root action ranked by mean return. After execution, we update the root from observation, revalidate reused candidates, discard inconsistent descendants, and reset search statistics.

Acknowledgments

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